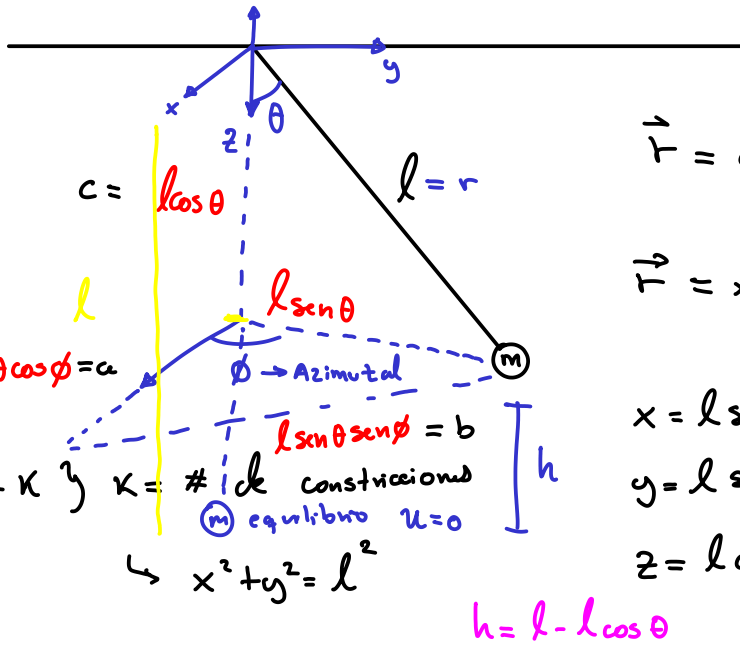




$$\vec{r} = a\hat{x} + b\hat{y} + c\hat{z}$$

$$\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$$

$$\left. \begin{aligned} x &= l \sin \theta \cos \phi \\ y &= l \sin \theta \sin \phi \\ z &= l \cos \theta \end{aligned} \right\} \vec{r} = l \sin \theta \cos \phi \hat{x} + l \sin \theta \sin \phi \hat{y} + l \cos \theta \hat{z}$$

$$\begin{aligned} gdl &= 3N - K \\ &= 3(1) - 1 = 2 \text{ gdl} \rightarrow 2 \text{ coordenadas independientes} \end{aligned}$$

Energía cinética

$$T = \frac{1}{2} m (|\dot{\vec{r}}|^2) = \frac{m}{2} (\dot{\vec{r}} \cdot \dot{\vec{r}}) = \frac{m}{2} (\dot{l}^2 + l^2 \dot{\theta}^2 + l^2 \sin^2 \theta \dot{\phi}^2)$$

Energía potencial

$$U = mgz \Rightarrow mg(l \cos \theta) \Rightarrow U = mg(l - l \cos \theta) \quad \text{constante...}$$

$$\begin{aligned} \text{Lagrangiana} \Rightarrow L &= L(\theta, \phi, \dot{\theta}, \dot{\phi}, t) = \frac{m}{2} (\dot{l}^2 + l^2 \dot{\theta}^2 + l^2 \sin^2 \theta \dot{\phi}^2) - mg(l(1 - \cos \theta)) \\ L &= T - U \\ &= \frac{m}{2} (l^2 \dot{\theta}^2 + l^2 \sin^2 \theta \dot{\phi}^2) - mg(l(1 - \cos \theta)) \end{aligned}$$

Recordando... $l = \text{cte}$

$$\begin{aligned} \dot{\vec{r}} &= [l \dot{\theta} \cos \theta \cos \phi - l \dot{\phi} \sin \theta \sin \phi] \hat{x} + [l \dot{\theta} \cos \theta \sin \phi + l \dot{\phi} \sin \theta \cos \phi] \hat{y} \\ &\quad - l \dot{\theta} \sin \theta \hat{z} ; |\dot{\vec{r}}|^2 = \dot{\vec{r}} \cdot \dot{\vec{r}} = [l \dot{\theta} \cos \theta \cos \phi - l \dot{\phi} \sin \theta \sin \phi]^2 + [l \dot{\theta} \cos \theta \sin \phi + l \dot{\phi} \sin \theta \cos \phi]^2 + [l \dot{\theta} \sin \theta]^2 \end{aligned}$$

$$+ l \dot{\phi} \sin \theta \cos \phi]^2 + l^2 \dot{\theta}^2 \sin^2 \theta$$

Se puede hacer el desarrollo

y llegamos a:

$$l^2 \dot{\theta}^2 + l^2 \sin^2 \theta \dot{\phi}^2$$

Teníamos...

$$-mgl + mgl \cos \theta$$

$$L(\theta, \phi, \dot{\theta}, \dot{\phi}, t) = \frac{m}{2} (l^2 \dot{\theta}^2 + l^2 \sin^2 \theta \dot{\phi}^2) - mgl(1 - \cos \theta)$$

$$q_1 = \theta \quad \left. \begin{array}{l} \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) - \frac{\partial L}{\partial \theta} = 0 \Rightarrow \frac{d}{dt} (m l^2 \dot{\theta}) - \left[\cancel{2} l^2 \sin \theta \cos \theta \dot{\phi}^2 - mgl \sin \theta \right] \\ q_2 = \phi \end{array} \right\}$$

$$\Rightarrow m l^2 \ddot{\theta} - \cancel{2} l^2 \sin \theta \cos \theta \dot{\phi}^2 + mgl \sin \theta = 0$$

$$\Rightarrow l^2 \ddot{\theta} - l^2 \sin \theta \cos \theta \dot{\phi}^2 + gl \sin \theta = 0$$

$$\Rightarrow \ddot{\theta} - \sin \theta \cos \theta \dot{\phi}^2 + \frac{g}{l} \sin \theta = 0$$

$$\Rightarrow \boxed{\ddot{\theta} = -\frac{g}{l} \sin \theta + \sin \theta \cos \theta \dot{\phi}^2}$$

$$\phi: \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}} \right) - \frac{\partial L}{\partial \phi} = 0 \Rightarrow \frac{d}{dt} (m l^2 \sin^2 \theta \dot{\phi}) = 0 \Rightarrow m l^2 \sin^2 \theta \dot{\phi} = \text{cte en el tiempo}$$

$l_z = \text{momento angular se conserva.}$

$$\rightarrow m l^2 \frac{d}{dt} (\sin^2 \theta \dot{\phi}) \quad \left[l_z = c \right] \Rightarrow \phi = \text{coordenada cíclica}$$

Ley de conservación

$$m l^2 [\sin^2 \theta \ddot{\phi} + 2 \dot{\theta} \dot{\phi} \sin \theta \cos \theta] = 0$$

Tenemos las siguientes ecuaciones de movimiento...

$$\ddot{\theta} = -\frac{g}{l} \sin \theta + \sin \theta \cos \theta \dot{\phi}^2$$

$$\dot{\phi} + 2\dot{\theta}\dot{\phi} \frac{\cos \theta}{\sin \theta} = 0 \Rightarrow \ddot{\phi} = -2\dot{\theta}\dot{\phi} \frac{\cos \theta}{\sin \theta}$$

Para hallar la Hamiltoniana del sistema aplicamos una transformada de Legendre a la función lagrangiana. Tenemos..

• La Hamiltoniana no depende de las velocidades.

$$H(\theta, \phi, p_\theta, p_\phi, t) = \sum_{i=1}^n \frac{\partial L}{\partial \dot{q}_i} \dot{q}_i - L(\theta, \phi, \dot{\theta}, \dot{\phi}, t)$$

$$p_\theta = ml^2 \dot{\theta}$$

$$p_\phi = ml^2 \sin^2 \theta \dot{\phi}$$

$$\dot{\theta} = \frac{p_\theta}{ml^2}$$

$$\dot{\phi} = \frac{p_\phi}{ml^2 \sin^2 \theta}$$

$$= \frac{\partial L}{\partial \dot{\theta}} \dot{\theta} + \frac{\partial L}{\partial \dot{\phi}} \dot{\phi} - L(\theta, \phi, \dot{\theta}, \dot{\phi}, t)$$

$$= ml^2 \dot{\theta}^2 + ml^2 \sin^2 \theta \dot{\phi}^2 - \left[\frac{m}{2} (l^2 \dot{\theta}^2 + l^2 \sin^2 \theta \dot{\phi}^2) - mgl(1 - \cos \theta) \right]$$

$$= ml^2 \left[\frac{p_\theta^2}{m^2 l^4} \right] + ml^2 \sin^2 \theta \left[\frac{p_\phi^2}{m^2 l^4 \sin^4 \theta} \right] - \frac{m}{2} l^2 \left[\frac{p_\theta^2}{m^2 l^4} \right] - \frac{m}{2} l^2 \sin^2 \theta \left[\frac{p_\phi^2}{m^2 l^4 \sin^4 \theta} \right]$$

$$+ mgl - mgl \cos \theta$$

$$= \frac{2}{2} \frac{p_\theta^2}{ml^2} + \frac{2}{2} \frac{p_\phi^2}{ml^2 \sin^2 \theta} - \frac{p_\theta^2}{2ml^2} - \frac{p_\phi^2}{2ml^2 \sin^2 \theta} + mgl - mgl \cos \theta$$

$$H(\theta, p_\theta, p_\phi, t) = \frac{p_\theta^2}{2ml^2} + \frac{p_\phi^2}{2ml^2 \sin^2 \theta} + mgl(1 - \cos \theta)$$

$$E = T + V = H$$

$$\hookrightarrow \left(\frac{p_\phi^2}{2ml^2} \sin^{-2} \theta \right) = - \frac{p_\phi^2}{ml^2} \sin^{-3} \theta \cos \theta$$

Tenemos Ecuaciones canónicas de Hamilton:

$$\left. \begin{aligned} \dot{\theta} &= -\frac{\partial H}{\partial \theta} ; & \dot{\theta} &= \frac{\partial H}{\partial p_{\theta}} \\ \dot{\phi} &= -\frac{\partial H}{\partial \phi} ; & \dot{\phi} &= \frac{\partial H}{\partial p_{\phi}} \end{aligned} \right\} \begin{aligned} \dot{\theta} &= -\frac{p_{\phi}^2}{ml^2 \sin^3 \theta} \cos \theta - mgl \sin \theta & \dot{\theta} &= \frac{p_{\theta}}{ml^2} \\ \dot{\phi} &= -[0] = 0 ; & \dot{\phi} &= \frac{p_{\phi}}{ml^2 \sin^2 \theta} \\ \phi &= \text{coordenada} \\ & \quad \underline{\text{cíclica}} \end{aligned}$$

$\Rightarrow p_{\phi} = \text{cte}$ (conservación del momento angular alrededor del eje vertical).